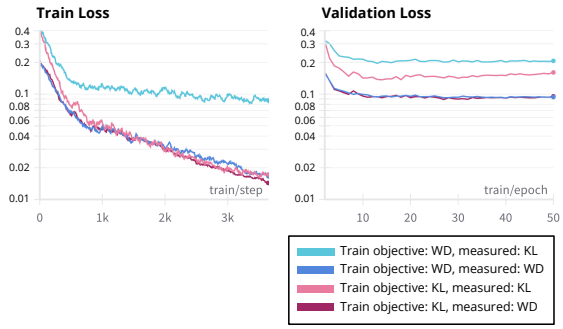

Figure 9: Train loss curve (left) and validation loss curve (right) for Llama-2-7B fine-tuned on 90% of OpinionQA, with the remaining 10% used for validation. Light and dark blue lines represent KL-divergence (KL) and Wasserstein distance (WD) when used KL as a training objective, while light and dark red lines represent KL and WD when used WD as a training objective. The two training objectives yield similar results in terms of WD, the primary measure of opinion distribution matching in our work.

WD converged to similar levels regardless of the objective (see Figure 9). We attribute this to KL-divergence penalizing low-probability assignments without significantly altering the overall distribution geometry. Given the KL divergence's broader applicability—without requiring ordinal information—we primarily used KL-divergence in our experiments.

C Additional Experiments

C.1 Effect of Response Distribution Modeling

In this section, we compare different methods for capturing the distribution of human responses. We consider three approaches:

1. *One-hot*: Predicting only the most probable response, which ignores the full distribution over all responses (Li et al., 2024).

2. *Augment by N*: Augmenting the dataset by replicating each response by a factor of N according to its observed frequency (Zhao et al., 2023).

3. *Explicit probability modeling*: Directly modeling the full response distribution using the actual probability values for each option.

Table 5 summarizes the results of these approaches. Notably, explicit probability modeling substantially outperforms the one-hot method, demonstrating that simply predicting the single most frequent response fails to capture the opinion diversity present within each subpopulation.

Compared with augment by N (2nd and 3rd column in Table 5), explicit probability modeling also achieves better performance. Importantly, the performance gap exceeds the quantization error introduced by discretizing the response distribution. For instance, when discretizing with a factor of N, the quantization error is $\frac{1}{2N}$—approximately 0.01 or 0.005 in the cases shown in Table 5. Moreover, explicit modeling offers the practical benefit of reducing the data volume by a factor of N compared to the augmentation approach, thereby lowering the computational cost of fine-tuning LLMs.

These results underscore the importance of explicit distribution modeling. By aligning the model's predictive distribution directly with the survey distribution, we achieve higher accuracy with fewer data samples, avoiding the rounding errors and replication overheads that are inherent to data-augmentation approaches.

C.2 Post-trained Model

We fine-tune Llama-2-7B-chat to observe the effect of starting from checkpoints that have been instruction-tuned via Reinforcement Learning from Human Feedback (RLHF). Table 6 shows the evaluation performance of a baseline method (Zero-shot prompt (`QA`)), fine-tuned base model (Llama-2-7B) and fine-tuned chat model (Llama-2-7B-chat). We observe the significant performance improvement, while the baseline method performs worse then the models not instruction-tuned (Table 1). Especially, the performance for `SubPOP`-Eval of chat model is significantly worse than that of base model. We observe the high WD of the baseline method resulting from the model assigning high probability to a specific token (e.g. 'A'), being far apart from the human opinion distribution. After fine-tuning the model are able to generate a more distributed probability of answer tokens. This result coincides with the result reported in (Moon et al., 2024).

Table 5: Comparison of evaluation performance for three response distribution modeling approaches, with Llama-2-7B as a base model. The last column (Explicit) is identical to the ours presented in Table 1. A model fine-tuned to predict the most probable choice (one-hot) performs the worst, as the model has not learned distributional opinion at fine-tuning phase. A model trained on augmented data (Aug. ($\times$50, $\times$100)), while performing much better than one-hot still underperforms the explicit distribution modeling.

Eval Dataset	One-hot	Aug. ($\times$ 50)	Aug. ($\times$ 100)	Explicit (Ours)
OpinionQA	0.163	0.110	0.107	0.106
`SubPOP`-Eval	0.178	0.130	0.123	0.121

Table 6: Performance of the fine-tuned Llama-2-7B-chat model (Chat LLM FT). For comparison, we also present lower and upper bounds, the baseline method Zero-shot prompt (QA) and our fine-tuned Llama-2-7B (Base LLM FT).

Method	OpinionQA	SubPOP-Eval
Upper bound (Unif.)	0.178	0.208
Lower bound (Human)	0.031	0.033
Base zero-shot prompt (QA)	0.173	0.206
Base LLM FT	0.106	0.121
Chat zero-shot prompt (QA)	0.308	0.383
Chat LLM FT	0.109	0.148

C.3 Generalization to Unseen Subpopulations

Here we present a complete list of evaluation performance on OpinionQA for unseen subpopulations (the groups not used to fine-tune our model) and perform an analysis that shows our fine-tuned models are able to steer towards the given subpopulation information.

As shown in Table 7, we observe a performance improvement across unseen subpopulations. To verify that the performance improvements arise from the fine-tuned model being able to steer towards given subpopulations, we measure *inter-group disagreement* pattern for the demographic and ideology traits, shown in Figure 10, 11, 12, and 13. We consistently observe across traits that the disagreement pattern of our model resembles that of the human group, while zero-shot prompting with the base model exhibits a pattern completely different from the human group result. This observation shows that our fine-tuned model learns to condition on subpopulation information and also generalizes to subpopulations unseen during fine-tuning.

D Baseline Details

- **Zero-shot prompting**: Three prompt styles—QA, BIO, and PORTRAY—are introduced in (Santurkar et al., 2023) to integrate group information into prompts. These prompts are then combined with survey questions to construct inputs for LLM. Then, the first-token log-probability from LLM is measured to calculate the model's response distribution over options. In our baseline (and also in fine-tuning experiments) we focus on the QA steering format. Examples of this prompting method are shown in Figure 14.

- **Few-shot prompting**: We craft a conditioning prompt that contains not only group information but also the group's response distribution to k train questions, following (Hwang et al., 2023). For a test question $q_{test} \in Q_{test}$, we first sort training questions Q_{train} into $\{q_1, q_2, ...\}$ such that $\texttt{sim}(\texttt{E}(q_1), \texttt{E}(q_{test})) > \texttt{sim}(\texttt{E}(q_2), \texttt{E}(q_{test}))$, and so on. $\texttt{E}(q)$ denotes the embedding model (OpenAI-text-embedding-3-large) output of the input q and sim is a cosine similarity between two embedding vectors. Then, response information of the first k questions $\{q_i, p(\mathcal{A}_{q_i}|q_i, g)\}_{i=1}^{k}$ are used as few shot prompts to have the language model verbalize (Meister et al., 2024) expected response distribution for the given g and q_{test}. An example of the prompt for $k=3$ case is shown in Figure 15, while we run the baseline experiment in a $k=5$ setting.

- **Modular Pluralism**: The intuition behind Modular Pluralism (Feng et al., 2024) is that a language model trained on a text corpus of a specific subpopulation will faithfully represent public opinion of that population. Given a survey question with a PORTRAY-style steering prompt, each of language model 'modules' (fine-tuned Mistral-7B-Instruct-v0.1) generates an option choice with explanation. A black-box LLM (GPT-3.5-turbo-Instruct) receives all generations and select a generation that best aligns with the given group. Finally, using the chosen generation as a context, a black-box LLM generates probability distribution over options. The example pipeline is shown in Figure 16. Instead of the sub-sampled OpinionQA dataset the authors of the method used, we use the exactly same evaluation set across all baseline methods and our approach for a fair comparison.

- **Upper bound**: We estimate the distribution between human responses and uniform distribution as an upper bound of WD metrics.

- **Lower bound**: We compute a lower bound by randomly sampling a group of respondents and calculating the Wasserstein distance (WD) between the distribution of the sampled group and that of the original respondents for each question. We then bootstrap with $R=1000$ to construct a 95% confidence interval (CI) for the WDs. Further details on this estimation process are provided below.

  Computing weighted answer distributions: For each group g and question q, we have n_{gq} responses from respondents who belong to group g answering question q: $x_1, x_2, \cdots, x_{n_{gq}}$, where $x_i \in \mathcal{A}_q$, i.e., the answer set for question

Table 7: Evaluation performance of our fine-tuned Llama-2-7B model on OpinionQA for subpopulations not included in the fine-tuning dataset SubPOP-Train. For reference, we present a lower bound (human) and the zero-shot prompting (QA). Absolute difference refers to the WD difference between zero-shot prompting and ours, and the relative improvement is calculated in a same way as Figure 3.

Attribute	Group	Lower Bound (Human)	Zero-shot (QA)	Ours	Absolute Diff.	Relative Improvement
Age	18-29	0.023	0.185	0.096	0.089	0.548
Age	30-49	0.014	0.151	0.093	0.058	0.424
Age	50-64	0.014	0.154	0.101	0.052	0.377
Age	65+	0.013	0.195	0.115	0.080	0.438
Region	Midwest	0.016	0.153	0.095	0.058	0.425
Region	West	0.017	0.162	0.095	0.068	0.465
Education	Associate's Degree	0.026	0.159	0.098	0.061	0.455
Education	High School Graduate	0.017	0.144	0.092	0.053	0.413
Education	Postgraduate	0.015	0.174	0.106	0.068	0.426
Education	Some College, No Degree	0.018	0.144	0.093	0.051	0.405
Income	$50,000-$75,000	0.016	0.153	0.098	0.054	0.396
Income	$30,000-$50,000	0.019	0.144	0.094	0.050	0.400
Political Ideology	Very Conservative	0.026	0.208	0.107	0.101	0.555
Political Ideology	Very Liberal	0.025	0.202	0.111	0.091	0.514
Political Party	Independent	0.016	0.155	0.093	0.062	0.445
Political Party	Something Else	0.026	0.162	0.092	0.069	0.510
Race	Other	0.050	0.180	0.144	0.036	0.275
Religion	Agnostic	0.028	0.189	0.115	0.074	0.459
Religion	Buddhist	0.063	0.207	0.149	0.059	0.405
Religion	Nothing in Particular	0.019	0.153	0.092	0.061	0.454
Religion	Orthodox	0.083	0.221	0.180	0.041	0.298
Religion	Other	0.051	0.184	0.123	0.061	0.457
Religion	Roman Catholic	0.018	0.145	0.098	0.047	0.371

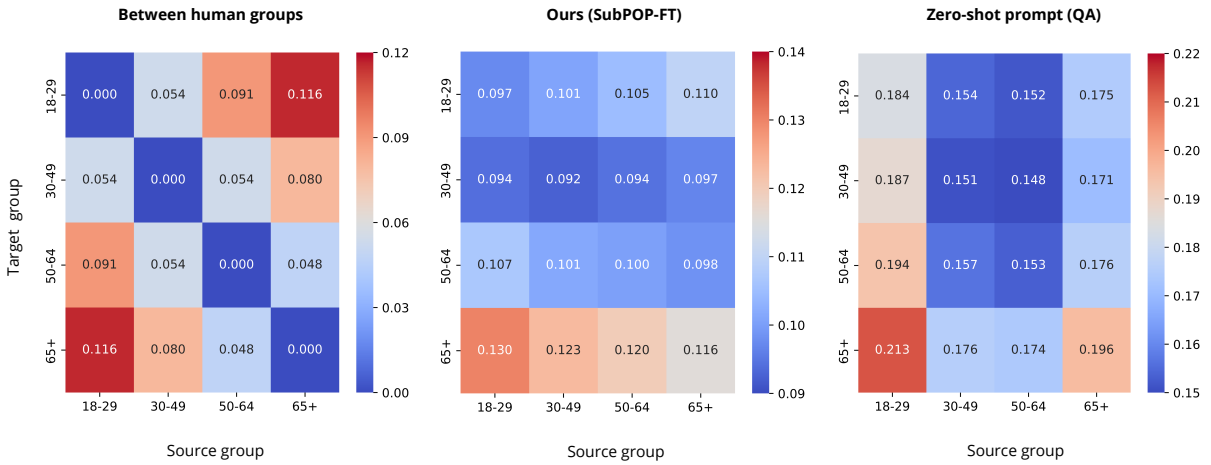

Figure 10: Heatmap of intergroup disagreement between a target human group (y-axis) and a source group (x-axis, either a human group or a group simulated with the language model), for OpinionQA evaluation data and age trait using Llama-2-7B as a base model. All subpopulations are unseen during fine-tuning.

q (e.g., $\{1,2,3,4\}$). Furthermore, each respondent (and thus, their response) is associated with a wave-specific weight $w_1, w_2, \cdots, w_{n_{gq}}$, provided by Pew Research. We compute the human answer distribution $\pi_{gq}^{(H)}$ as a weighted sum over responses, where the proportion of respondents providing answer $a \in \mathcal{A}_q$ is estimated as

$$\pi_{gq}^{(H)}(a) = \frac{\sum_{i=1}^{n_{gq}} w_i \mathbb{1}[x_i = a]}{\sum_{i=1}^{n_{gq}} w_i}.$$

Bootstrapping at the respondent-level: We draw bootstrap samples per group at the respondent-level including questions from all survey waves. This allows us to capture correlations in answer distributions across questions and across waves.

Specifically, let $\mathcal{P}_g$ represent the set of respondents in group g, where $|\mathcal{P}_g| = n_g$. We produce bootstrapped samples by repeatedly sampling n_g respondents from $\mathcal{P}_g$ with replacement. Let $p_1^{(r)}, p_2^{(r)}, \cdots, p_{n_g}^{(r)}$ represent the sampled respondents for the r-th bootstrap, and let $w_1^{(r)}, w_2^{(r)}, \cdots, w_{n_g}^{(r)}$ represent their corresponding weights.

For each question q, let $\mathcal{P}_{gq} \subseteq \mathcal{P}_g$ represent the set of respondents from group g who answered question q; as before, $|\mathcal{P}_{gq}| = n_{gq}$. Let us define $q(p_i)$ as person p_i's response to question q if p_i answered question q, i.e., $p_i \in \mathcal{P}_{gq}$, and 0 otherwise. Then, we compute the r-th answer